When the Earth Moves

The ground under Peru moves every day. Read what really happened in 1746, 1868, 1970, 2001 and 2007, look at real photos and numbers, and learn what every family must have ready. Then make a safety flyer that can help the people in your street.

Inside this book

- 12 chapters, one for each week
- Activities after every chapter
- Peru in Photos
- Two timelines
- Glossary with Spanish
- How am I doing? rubric

Movers edition · A1

Nordic International School · Grade 3 · Unit 5 · Lima, Peru

Photos inside!

NORDIC INTERNATIONAL SCHOOL · GRADE 3 · UNIT 5

When the Earth Moves

Earthquakes, Tsunamis and Volcanoes in Peru

Movers edition · A1

Fun for Nordic · The Fjord Club

Before you start

BIG QUESTION

Peru shakes. What do I have to know, and what do I have to have ready, so that it does not catch my family by surprise?

Peru shakes. Almost every day, somewhere in the country, the ground moves a little. Most of the time nobody notices. But five times in its history, an earthquake has changed Peru forever: in 1746, 1868, 1970, 2001 and 2007.

In this book you will read what really happened, with real numbers. Then you will use what you learn to make something useful: a **safety flyer** that a family in your street could follow without asking anybody.

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Meet The Fjord Club

Valentina

tells the stories

Erik

explains the grammar with his inventions

Sofía

loves maps and places

Mateo

always forgets something

Luna

the husky, only barks for good answers

Contents

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How am I doing?

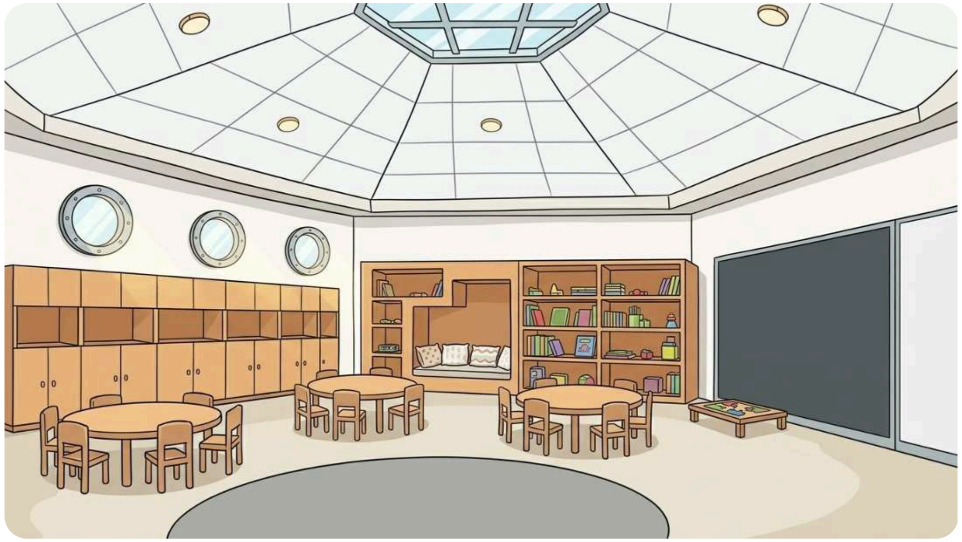
The unit rubric. AD is the highest level.

I can...	AD	A	B	C
I name natural disasters, safety words, the things in the emergency kit and the parts of the body.	I use them and I explain what one of them means to somebody who does not know it.	I use the words on my own, in sentences, when I am talking about safety.	I say most of the words when I see the picture.	I say a few of the words with help.
I read a table and a chart about earthquakes in Peru and I say what they show.	I notice something in the data that nobody pointed out to me.	I ask and answer questions about the data and I compare two earthquakes.	I find information when I am asked a direct question.	I find one piece of information if somebody shows me the column.
I tell what happened in a real Peruvian earthquake using the past simple.	I tell it in order and I add what happened afterwards.	I tell what happened in four sentences, in order and in the past.	I write two or three sentences in the past with some mistakes.	I copy sentences in the past.
I use must, must not and should to give safety rules and advice.	I explain to somebody else why one is a rule and the other is advice.	I choose must, must not or should correctly and I give the reason.	I write rules but I mix up must and should.	I copy a rule that is given to me.
I organise safety instructions into before, during and after.	I notice when an instruction could go in two moments and I say why.	I sort them all and I use the three moments to organise my own flyer.	I sort most of the actions with a bit of help.	I put two of the three moments in the right place.
I write a safety flyer of 90 to 110 words and I present it to other people.	My flyer works for somebody who is younger than me, and I prove it.	I write the three parts, with a heading, an instruction and a drawing each, and I present it.	I write the three parts but they are short or unclear.	I write some sentences and I show my drawing.

CHAPTER 1

W1 · Lanzamiento

Our Ground Moves



A normal Monday morning at school — until the windows start to rattle.

Look at the map! Peru is on the Ring of Fire.

It was a normal Monday morning at school in Lima. The children were reading quietly. Suddenly the windows started to rattle. The lamp above the teacher's desk moved from side to side, and the floor shook under their feet.

“Is it a big truck?” asked Mateo.

It wasn't a truck. It was a small earthquake. “Stay calm,” said the

first-aid kit *(botiquín)* — a box with plasters, bandages and things to help people who are hurt

flood *(inundación)* — a lot of water covering land that is usually dry

have to *(tener que)* — it is necessary: we have to leave now

head *(cabeza)* — the top part of your body, with your eyes and mouth

headache *(dolor de cabeza)* — a pain in your head

hurt *(doler / lastimarse)* — to feel pain: my knee hurts

intensity *(intensidad)* — how strongly the shaking is felt in one place

knee *(rodilla)* — the middle of your leg, where it bends

landslide *(deslizamiento / huaico)* — rocks and earth sliding down a hill

lava *(lava)* — hot liquid rock that comes out of a volcano

magnitude *(magnitud)* — a number that shows how strong an earthquake is

meeting point *(punto de reunión)* — the safe place outside where everyone meets after leaving

must / must not *(debe / no debe)* — for rules: you must pack water; you must not run

plate *(placa tectónica)* — a giant piece of the Earth's outside

Ring of Fire *(Cinturón de Fuego)* — the line of coasts around the Pacific with many earthquakes and volcanoes

safe zone *(zona segura)* — the safest place to be in an earthquake

seismograph *(sismógrafo)* — a machine that measures the shaking of the ground

shelter *(refugio / albergue)* — a safe place to stay

should *(debería)* — for advice: you should check your backpack

stomachache *(dolor de estómago)* — a pain in your stomach

torch *(linterna)* — a small light you carry in your hand

tsunami *(tsunami / maremoto)* — very big waves caused by an earthquake under the sea

volcano *(volcán)* — a mountain where hot rock, gas and ash come out

whistle *(silbato)* — a small thing you blow into to make a loud sound

Glossary

The words of this unit, with Spanish.

after *(después)* — at a later time: after the earthquake

aftershock *(réplica)* — a smaller earthquake that comes after a big one

ankle *(tobillo)* — the part of the body between your leg and your foot

ash *(ceniza)* — the grey powder that comes out of a volcano

avalanche *(avalancha / aluvión)* — a very large amount of ice, snow and rock falling fast down a mountain

back *(espalda)* — the part of your body behind you, from your neck to your bottom

backache *(dolor de espalda)* — a pain in your back

before *(antes)* — at an earlier time: before the earthquake

blanket *(manta)* — a warm cover for your bed or for sleeping outside

bruise *(moretón)* — a blue or purple mark on your skin after a knock

cut *(corte)* — an opening in your skin that bleeds

danger *(peligro)* — something that can hurt you

drill *(simulacro)* — practice for what to do in an emergency

Drop, cover, hold on *(Agáchate, cúbrete, sujétate)* — what you do during the shaking

during *(durante)* — while something is happening

earthquake *(sismo / temblor / terremoto)* — a sudden shaking of the ground, big or small: in English it is always "earthquake"

emergency kit *(mochila para emergencias)* — a bag with the things you need in an emergency

epicentre *(epicentro)* — the place on the ground right above where an earthquake starts

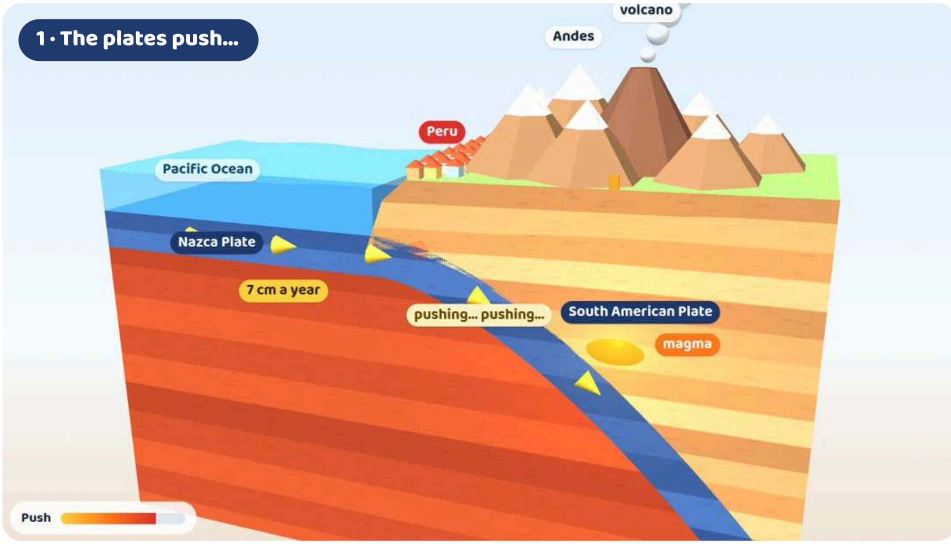
exit *(salida)* — the way out of a building

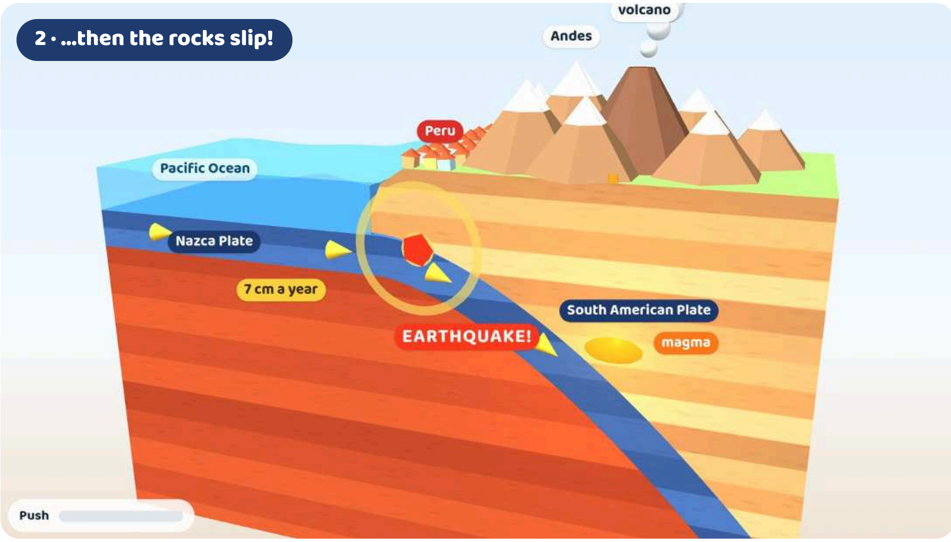
fault *(falla)* — a long crack in the rock where the ground can suddenly move

teacher. After a few seconds, everything was quiet again. Nobody was hurt. In Peru, this happens a lot.

A giant puzzle

The outside of the Earth is not one big piece. It is broken into giant pieces of rock called **plates**, like the pieces of a puzzle. The plates move very, very slowly.
Peru is on the **South American Plate**. Under the Pacific Ocean, next to Peru, there is another plate: the **Nazca Plate**.





The Nazca Plate slides under the South American Plate. The push builds up, then the rocks slip: an earthquake!

3D model — watch it move in the web edition.

The two plates push against each other. The Nazca Plate is heavier, so it goes down under Peru — about seven centimetres every year. That's about the length of your finger!

But the plates don't move smoothly. They get stuck. For many years they push and push, and the rocks bend. Then, suddenly, the rocks slip. The ground shakes. That is an **earthquake**.

Timeline of the World

1556 —An earthquake of about magnitude 8 in Shaanxi, China, is the deadliest in history: about 830,000 people die.

1755 —An earthquake of about 8.5, a tsunami and fires destroy Lisbon, Portugal, and lead to scientific studies of earthquakes.

1906 —A magnitude 7.9 earthquake and fires destroy much of San Francisco, USA.

1912 —Alfred Wegener suggests that the continents move.

1935 —Charles Richter and Beno Gutenberg invent a scale to measure the size of earthquakes.

1960 —Valdivia, Chile: magnitude 9.5, the most powerful earthquake ever recorded. Its tsunami reaches Hawaii and Japan.

1964 —A magnitude 9.2 earthquake in Alaska, the biggest in North America.

2004 —The magnitude 9.1 Sumatra earthquake causes the deadliest tsunami ever recorded, around the Indian Ocean.

2010 —A magnitude 8.8 earthquake in Maule, Chile; its tsunami also reaches the coast of Peru.

2011 —A magnitude 9.1 earthquake hits Japan; the tsunami causes a nuclear accident at Fukushima.

2015 —A magnitude 7.8 earthquake in Nepal causes an avalanche at Everest Base Camp.

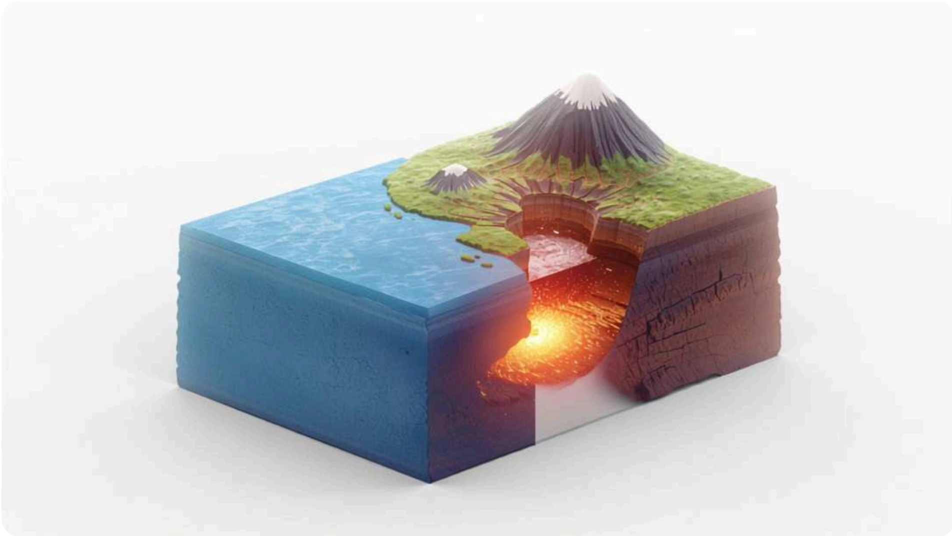
2023 —Two powerful earthquakes (7.8 and 7.5) hit Türkiye and Syria; about 59,000 people die.

2025 —A magnitude 8.8 earthquake in Kamchatka, Russia, sends tsunami alerts all around the Pacific — including Peru.

2019 —A magnitude 8.0 earthquake strikes deep under Loreto, in the Amazon.

2025 —The IGP reports about 840 earthquakes. After a magnitude 8.8 earthquake in Russia, Peru closes 65 ports as a tsunami precaution.

2026 —More than nine million schoolchildren take part in the first national drill of the year, on 29 May.



A slice of the Earth: the ocean plate goes down under the land.

The Ring of Fire

Peru is on the **Ring of Fire.** It is a very long line of coasts and islands around the Pacific Ocean, from Chile to Alaska and Japan, and down to New Zealand. Most of the world's earthquakes happen on the Ring of Fire. There are a lot of volcanoes there too.

The Ring of Fire goes all around the Pacific Ocean. Peru is on the right.

3D model — interactive in the web edition.

The Puzzle Man

In 1912, a German scientist called Alfred Wegener looked carefully at a map of the world. He noticed that South America and Africa fit together like two pieces of a puzzle. He said that long ago all the land was one giant piece, and that the continents move. Many scientists laughed at him. Wegener died in 1930, and nobody believed him. Many years later, in the 1960s, scientists found the proof. Wegener was right!

Photo: Unknown photographer · Public domain

Timeline of Peru's Earthquakes, Tsunamis and Volcanoes

1582 —An earthquake of about magnitude 8 destroys much of Arequipa.

1600 —Huaynaputina erupts — the biggest eruption in South America's recorded history. Ash covers Arequipa.

1604 —An earthquake and tsunami hit Arequipa and Arica; the waves flood about 1,200 km of coast.

1687 —An earthquake and tsunami damage Lima and Callao.

1746 —Lima–Callao earthquake (about 8.8). A 10-metre tsunami destroys Callao; only about 200 people there survive.

1868 —Arica earthquake (about 8.8). A 16-metre tsunami carries the USS Wateree 430 metres inland and reaches Hawaii and New Zealand.

1922 —An observatory opens in Huancayo: the beginning of the Instituto Geofísico del Perú.

1940 —A magnitude 8.2 earthquake shakes Lima; 179 people die.

1970 —Áncash earthquake (7.9). The Huascarán avalanche buries Yungay; about 70,000 people die in the region.

1972 —Peru creates INDECI, its national civil defence institute.

1974 —A magnitude 8.1 earthquake hits Lima and Callao, with a small tsunami at La Punta.

2001 —Southern Peru earthquake (8.4). A tsunami reaches Camaná and a tower of Arequipa cathedral falls.

2007 —Pisco earthquake (8.0): 596 people die and thousands of adobe houses fall.

2016 —Sabancaya volcano begins erupting on 6 November — and is still erupting today.

Students in Lima practise an earthquake drill.

Two every day

Scientists at the **IGP** (Instituto Geofísico del Perú) watch the ground all day and all night. In 2025, they wrote a report for about 840 earthquakes in Peru. That's more than two every day! Most of them are small, far under the sea or deep under the ground.

DID YOU KNOW?

Nobody in the world — not even the IGP — can say exactly when an earthquake will come. That is why every family must be ready.

The words for what the Earth does

earthquake
The ground moves and shakes, often for a few seconds.

tsunami
Very big, fast waves that can come after an earthquake under the sea.

landslide
Rocks and earth suddenly slide down a hill or mountain.

flood
A lot of water covers land that is usually dry.

Animals and Earthquakes

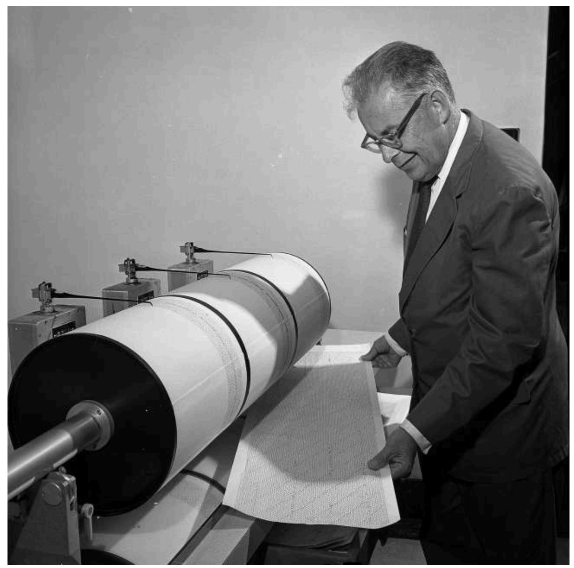
The ancient Greeks were some of the first people to notice that animals sometimes act strangely before earthquakes. Dogs bark, chickens stop laying eggs and bees leave their hives. But so far, there is no scientific proof for these stories.

In Peru, this sign shows a safe place to go in an earthquake.

Photo: AgainErick · CC BY-SA 3.0 (also GFDL)

This sign in Lima shows which way to go if a tsunami comes.

Photo: Gaboct · CC BY-SA 4.0



Charles Richter made a scale to tell how strong an earthquake is.

Photo: Gil Cooper, Los Angeles Times · CC BY 4.0

Alfred Wegener said that the continents move very slowly.

Photo: Unknown photographer · Public domain

Now you try!

A Your first page: the four words with a drawing each.

✋ Hand in · W1

Draw each word and write one sentence under it.

earthquake	tsunami
landslide	flood

B Match the words and the meanings.

1. plate	a. a line of coasts around the Pacific
2. Ring of Fire	b. a giant piece of the Earth's outside
3. IGP	c. the mountains of Peru
4. Andes	d. the scientists who watch the ground in Peru
5. Nazca Plate	e. the plate under the Pacific next to Peru

C True or false?

1. The Nazca Plate goes under Peru. **T / F**
2. The plates move very fast. **T / F**
3. In 2025, the IGP wrote reports for about 840 earthquakes. **T / F**
4. Alfred Wegener was from Peru. **T / F**
5. The Nazca Plate moves about seven **metres** every year. **T / F**

D Label the diagram of the plates.

Use the 3D picture of the plates. Write: **Nazca Plate · South American Plate · Pacific Ocean · Andes · earthquake**

1 ____________ 2 ____________ 3 ____________

4 ____________ 5 ____________

E Have you felt one? Ask five friends.

Ask: **"Have you ever felt an earthquake?" "Where were you?" "What did you do?"** Write the answers.

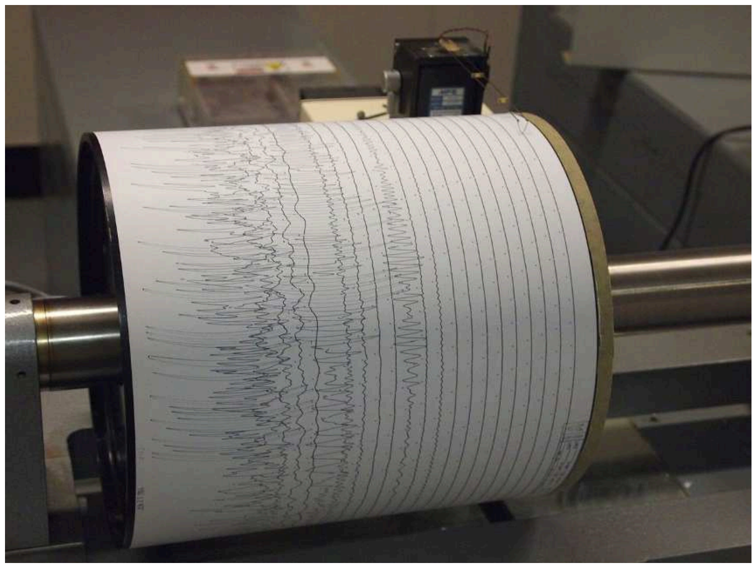
Name	Where were you?

F Maths: two every day

There are about 2 earthquakes every day in Peru. How many is that in a week (7 days)? ____ In 30 days? ____

In 2006, Ubinas volcano blew out a cloud of ash. This photo was taken from space.

Photo: NASA astronaut aboard the International Space Station · Public domain (NASA)



A seismograph draws wiggly lines when the ground shakes.

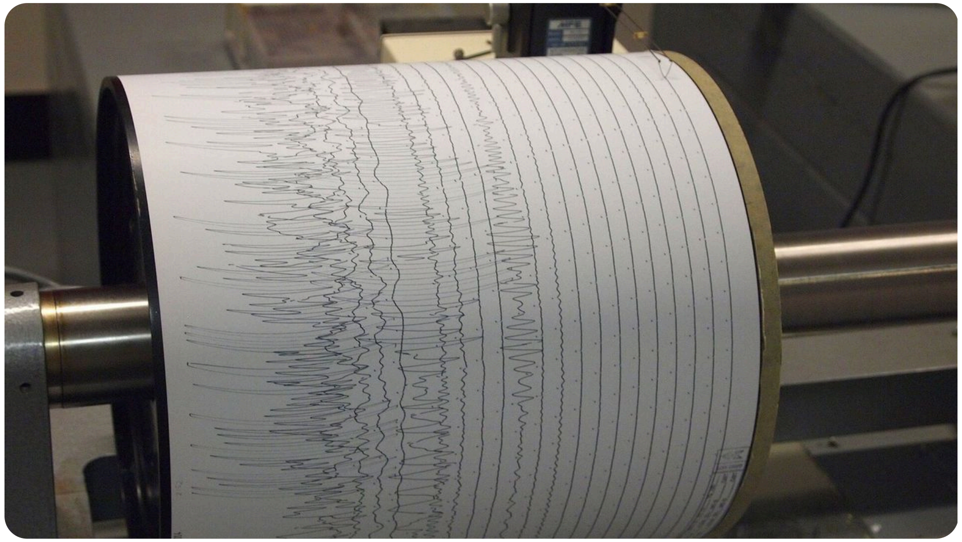
El Misti volcano stands over the city of Arequipa.

Photo: Josep M. Gracia · CC BY-SA 4.0

Sabancaya volcano sent up a big cloud of ash in 2017.

Photo: Ministerio de Defensa del Perú · CC BY 2.0

Reading the Numbers



A seismograph draws the shaking of the ground as a zig-zag line.

Photo: Z22 · CC BY-SA 3.0

A bigger number means a MUCH stronger earthquake.

In a quiet room in Lima, and in more than a hundred other places in Peru, special machines are listening to the ground. They are called **seismographs**. When the ground shakes, they draw a zig-zag line. A small earthquake draws small zig-zags. A big earthquake draws very big ones. The IGP uses these lines to find out where the earthquake was and how strong it was.

Richter Scale

There was no scale to measure the strength of earthquakes until 1935. That year, Charles F. Richter, a scientist in California, made a number scale with his friend Beno Gutenberg. Each whole number on the scale is ten times bigger than the number before it. So on a seismograph, an earthquake of magnitude 8 makes waves ten times bigger than one of magnitude 7! Today scientists use a newer scale, but it works in a similar way — and people still say "Richter".

Photo: Gil Cooper, Los Angeles Times · CC BY 4.0

Magnitude or Intensity?

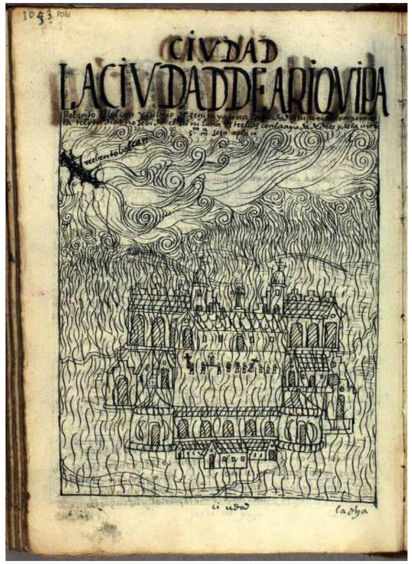
The **magnitude** is the energy of the earthquake. Every earthquake has only one magnitude: the Pisco earthquake was 8.0 in Pisco and 8.0 in Lima. The **intensity** is how strong the shaking was where *you* were, and it changes from place to place. It is strongest near the **epicentre** — the place on the ground right above where the earthquake started. After a big earthquake, smaller ones called **aftershocks** can come for days.

Five earthquakes Peru remembers

Peru has had thousands of earthquakes. But five of them changed the country. Read the table carefully. Every row is an earthquake. Every column tells you something about it.

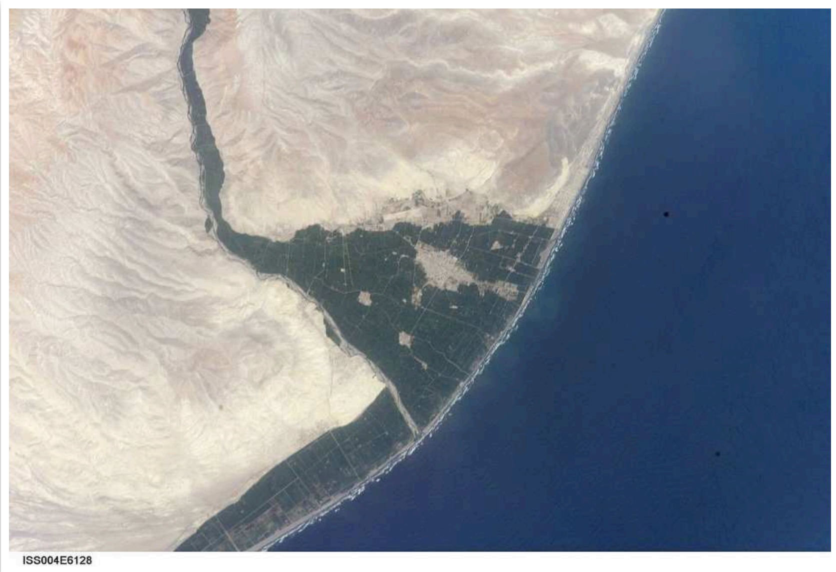
Helpers unloaded boxes of supplies for families after the 2007 earthquake.

Photo: Tech. Sgt. Jeremy Lock, U.S. Air Force · Public domain (U.S. military)



This old drawing shows ash falling on Arequipa when Huaynaputina erupted in 1600.

Photo: Felipe Guaman Poma de Ayala · Public domain



Astronauts took this photo of Camaná, where a tsunami came on land in 2001.

Photo: NASA Earth Sciences and Image Analysis Laboratory, Johnson Space Center · Public domain (NASA)

After the earthquake of 2007, broken bricks filled the streets of Pisco.

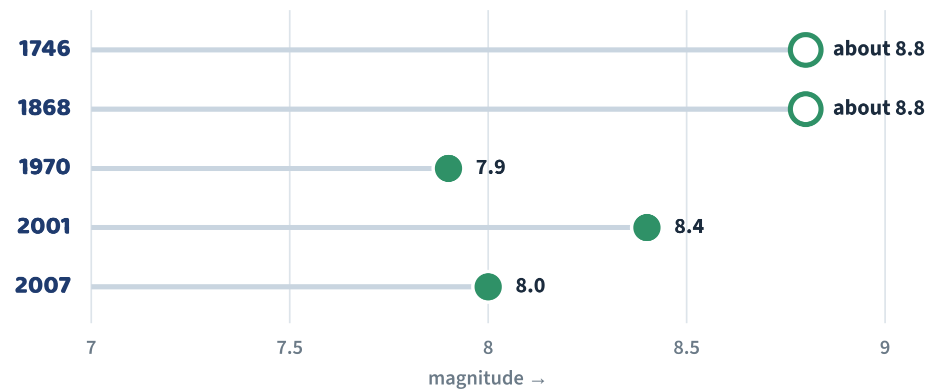
Photo: Senior Airman Shaun Emery, U.S. Air Force · Public domain (U.S. military)

Five earthquakes Peru remembers

Year	Where	Magnitude	Tsunami?	People who died
1746	Lima and Callao	about 8.8	Yes — about 10 m	about 6,000
1868	Arica (then in Peru)	about 8.8	Yes — about 16 m	thousands
1970	Áncash (Chimbote, Yungay)	7.9	No — an avalanche	about 70,000
2001	Arequipa, Moquegua, Tacna	8.4	Yes — at Camaná	77 (and 68 missing)
2007	Pisco, Ica, Chincha	8.0	Small	596

Magnitudes: USGS (IGP gives 7.9 for 2007). 1746 and 1868 are estimates — there were no seismographs then. Deaths: NOAA, INDECI.

Look at the table again. The 1746 and 1868 earthquakes were the strongest. But the earthquake that killed the most people was in 1970. Why? Because the shaking caused a giant avalanche. You will read about it in the next chapter.



The magnitude of the five earthquakes. Remember: one step on the scale makes waves ten times bigger!

The IGP: Peru's Earthquake Watchers

The Instituto Geofísico del Perú started in 1922 in Huancayo, in the Andes. Today its scientists watch earthquakes and volcanoes all over Peru, every day and every night. After an earthquake, they write a report in a few minutes. You can read it on their website!

DID YOU KNOW?

The biggest earthquake ever measured was in Valdivia, Chile, in 1960: magnitude 9.5. Its tsunami crossed the Pacific and reached Hawaii and Japan.

Asking about data

You can ask and answer questions about a table:

When was the earthquake in Arequipa? — It was in 2001.

Which earthquake was the strongest? — The 1746 and 1868 earthquakes.

How many people died in Pisco? — 596 people.

Was the 2001 earthquake **stronger than** the 2007 earthquake? — Yes, it was.

Today the white statue in Yungay looks out at Huascarán.

These palm trees grew in the main square of old Yungay. They are still there today.

After the avalanche of 1970, the statue on the cemetery hill in Yungay was still standing.

Photo: U.S. Geological Survey · Public domain (USGS)

In 1980, people built this gate at the Yungay memorial. Huascarán is behind it.

Photo: DB · Public domain (released by the author)

Now you try!

A Five true sentences taken from the table.

✋ Hand in · W2

Write five true sentences. Use the table. Example: *The Pisco earthquake was in 2007.*

B Read the table and answer.

1. When was the earthquake in Arequipa? ____________
2. Which earthquake had a 16-metre tsunami? ____________
3. How many people died in 2007? ____________
4. Which earthquakes were magnitude 8.8? ____________
5. Was there a tsunami in 1970? ____________

C Compare two earthquakes.

Complete with **stronger than · weaker than · more · fewer**.

1. The 1868 earthquake was ________ the 2007 earthquake.
2. The 1970 earthquake was ________ the 2001 earthquake.
3. ________ people died in 1970 than in 2007.
4. ________ people died in 2001 than in 2007.

D Read the chart.

Look at the magnitude chart. Which earthquake is the lowest on the scale? Which two are the highest? Write two sentences.

E Richter maths

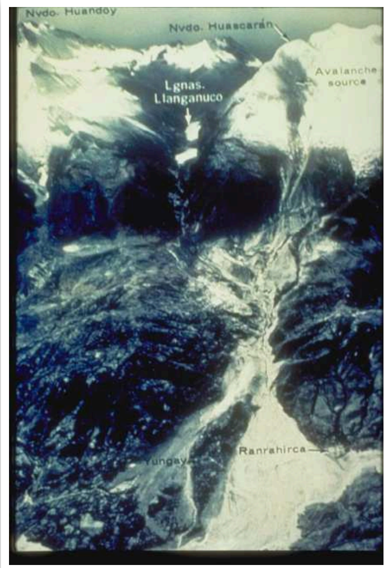
Each step on the scale makes the seismograph waves 10 times bigger. Magnitude 6 is 10 × magnitude 5. Magnitude 7 is ____ × magnitude 5. Magnitude 8 is ____ × magnitude 5.

F Speaking: data detectives

Work in pairs. Student A chooses an earthquake and answers questions. Student B asks: *When was it? Where was it? Was there a tsunami?* and guesses the earthquake.

Huascarán is the highest mountain in Peru.

Photo: Maurice Chédel · Public domain (released by the author)



From the sky you can see the path of the 1970 avalanche from Huascarán down to Yungay.

Photo: George Plafker, U.S. Geological Survey · Public domain (USGS / NOAA)

Parts of the Wateree are still on the beach near Arica today.

Photo: Pinot · Public domain (released by the author)

The Morro de Arica is a tall rock hill next to the sea.

Photo: Jowyto · CC BY-SA 3.0

CHAPTER 3

W3 · Construcción

What Happened in 1970

Before 1970: the town of Yungay in the valley below Huascarán, the highest mountain in Peru.

This is a true story. I will tell it in order — first, then, after that...

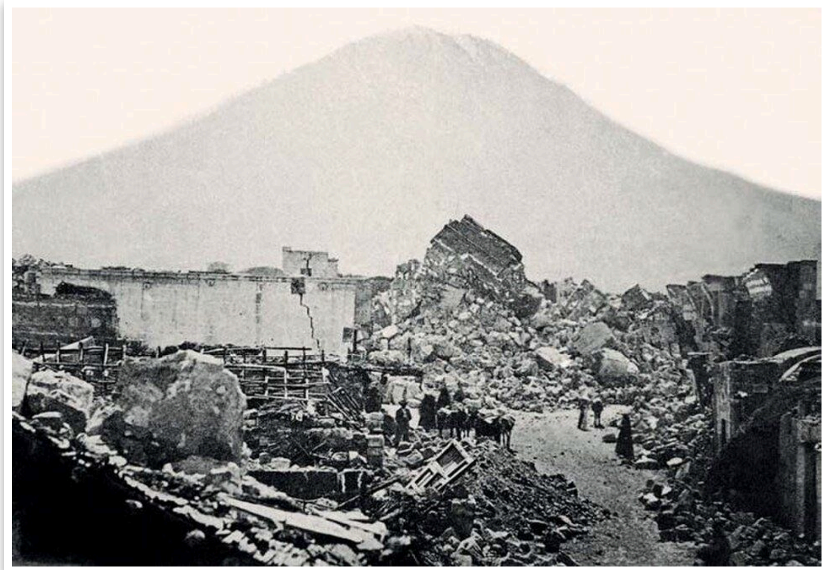
Sunday, 31 May 1970, was a beautiful sunny day in the region of Áncash, in the north of Peru. Above the valley stood **Huascarán**, the highest mountain in the country, covered in snow and ice. Below it, the town of Yungay was full of life. It was market day. The football World Cup started that afternoon in Mexico, and many families were at home watching it on TV. Hundreds of children were at a circus in the town's stadium.

At 3:23 p.m., an earthquake started under the sea, near the port of Chimbote. It was magnitude 7.9. The ground shook for about 45 seconds. In Huaraz, Chimbote and many other towns, houses made of adobe (mud bricks) fell down.

The shaking broke a giant piece of ice and rock off Huascarán.

Then something terrible happened. The shaking broke a giant piece of ice and rock off the top of Huascarán. It rushed down the mountain, mixed with snow, mud and rocks. It was an **avalanche**, and it travelled at about 300 kilometres an hour — faster than a racing car! In about three minutes it reached Yungay and covered the whole town, and most of the village of Ranrahirca too.

A few people saw the avalanche coming and ran up the cemetery hill, the highest place in the town. About 92 people climbed the hill and survived. The children at the circus were also on higher ground, and about 300 of them were saved. But around 20,000

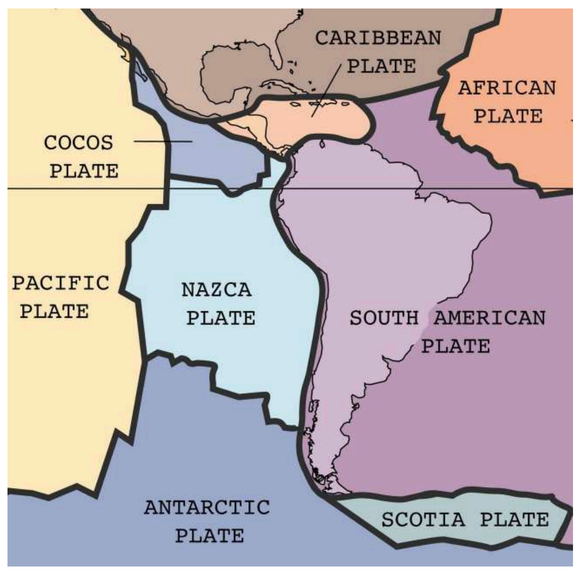


The earthquake of 1868 knocked down many buildings in Arequipa.

Photo: Unknown photographer, c. 1868 · Public domain

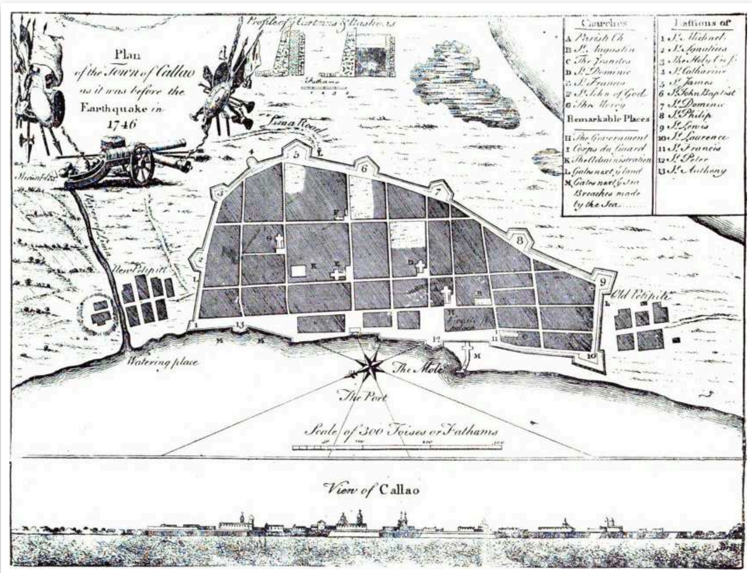
The tsunami of 1868 carried the ship Wateree onto the land at Arica.

Photo: Unknown photographer, 1868; Henry Clay Cochrane Collection, USMC Archives · CC BY 2.0



Peru is where the Nazca Plate and the South American Plate meet.

Photo: U.S. Geological Survey · Public domain (USGS)



This old map shows the port of Callao before the earthquake and tsunami of 1746.

Photo: Unknown 18th-century engraver; reproduced by The West Coast Leader, Ltd. · Public domain

people died in Yungay. In the whole region, about 70,000 people died. It was the worst disaster in the history of Peru.

The Four Palm Trees

After the avalanche, the only things you could see of Yungay's main square were the tops of four palm trees. Today those palm trees are still standing in the Campo Santo, the memorial park where old Yungay was. They are the symbol of the town.

Photo: Yhhue91 · CC BY-SA 4.0

Today old Yungay is a memorial park, the Campo Santo, with the statue on the cemetery hill and Huascarán behind.

After the earthquake

After the earthquake, people from all over Peru and from many other countries helped. They sent doctors, food, water, blankets and tents. Two years later, in 1972, Peru created **INDECI**, the office that helps the country prepare for disasters. And every year on 31 May, schools in Peru remember Yungay and practise what to do in an earthquake.

Erik's grammar machine

Past simple with -ed

start → start**ed** · rush → rush**ed** · climb → climb**ed** · cover → cover**ed** · help → help**ed** · destroy → destroy**ed**

was / were: It **was** a sunny day. The children **were** at a circus.

Irregular: fall → **fell** · run → **ran** · come → **came** · see → **saw**

Peru in Photos

Real photos of Peru's earthquakes, tsunamis and volcanoes — from 1746 to today.

Many old houses in Lima have wooden balconies.

Photo: Felipe Restrpo Acosta · CC BY-SA 4.0

Now you try!

A **Your presentation and the family reflection.** ✋ Hand in · W11

Present your flyer at the fair. Then write three or four honest lines: *My family already has… We don't have… This week I am going to…*

B Questions visitors might ask

Prepare answers: *Why must we drop, cover and hold on? Where is our meeting point? What is 119? What should I do near the sea?*

C How did I do? (Rubric)

Use the rubric at the back of the book. For each criterion, write AD, A, B or C and one piece of evidence.

Now you try!

A **Your four-sentence report about 1970.** ✋ Hand in · W3

Write four sentences in the past simple, in order. Use **First, Then, After that, In the end**.

B Put the events in order (1–6).

☐ The avalanche covered Yungay.
☐ Families watched the World Cup on TV.
☐ The earthquake started under the sea near Chimbote.
☐ A piece of ice and rock broke off Huascarán.
☐ Peru created INDECI.
☐ About 92 people ran up the cemetery hill.

C Write the verbs in the past.

1. The earthquake (start) ________ at 3:23.
2. The avalanche (rush) ________ down the mountain.
3. 92 people (climb) ________ the hill.
4. Other countries (help) ________ Peru.
5. It (be) ________ a sunny day.
6. The children (be) ________ at the circus.

D Then and now

Look at the photos of Yungay in 1970 and today (in the photo section). Write two sentences: *In 1970, … Today, …*

E A message of solidarity

31 May is the Day of Solidarity. Write a short message (2 sentences) for the people of Yungay today.

does not catch my family by surprise? Now you know the answer — and so does a family in your street.

🎤 Presenting my flyer

1. Hello! I'm … and this is my safety flyer.
2. It has three parts: before, during and after.
3. Before an earthquake, you must … .
4. During the shaking, … .
5. After the earthquake, … . Do you have any questions?

And in my house?

Now look at your own home. Be honest! What does your family already have ready? What is missing? What is **one thing** you are going to do about it this week?

At home	We have it ✓	We don't have it ✗
emergency backpack	☐	☐
water	☐	☐
torch and batteries	☐	☐
whistle	☐	☐
first-aid kit	☐	☐
a family meeting point	☐	☐
we know the safe zones at home	☐	☐
we know the 119 number	☐	☐
heavy things are on low shelves	☐	☐

Remember the big question at the start of this book? *Peru shakes. What do I have to know, and what do I have to have ready, so that it*

CHAPTER 4

W4 · Borrador

When the Sea Comes

In 1746 a tsunami destroyed the port of Callao.

If you are near the sea and the ground shakes a lot — go up!

On the night of 28 October 1746, at about half past ten, a very strong earthquake shook Lima. Many churches and houses fell. About half an hour later, the sea came. A giant wave, about ten metres high, crashed into the port of Callao. About 5,000 people lived in Callao. Only about 200 survived.

What is a tsunami?

When a big earthquake happens under the sea, the sea floor suddenly moves up or down. It pushes all the water above it, and very big waves start to travel. This is a **tsunami**. In the deep ocean the waves are as fast as a plane. Near the coast they become slower, but much taller. A tsunami is not one wave — many waves can come, for many hours.

> **A Japanese Word**
>
> "Tsunami" is a Japanese word: *tsu* means harbour and *nami* means wave. Japan has many tsunamis, like Peru, because it is on the Ring of Fire too.

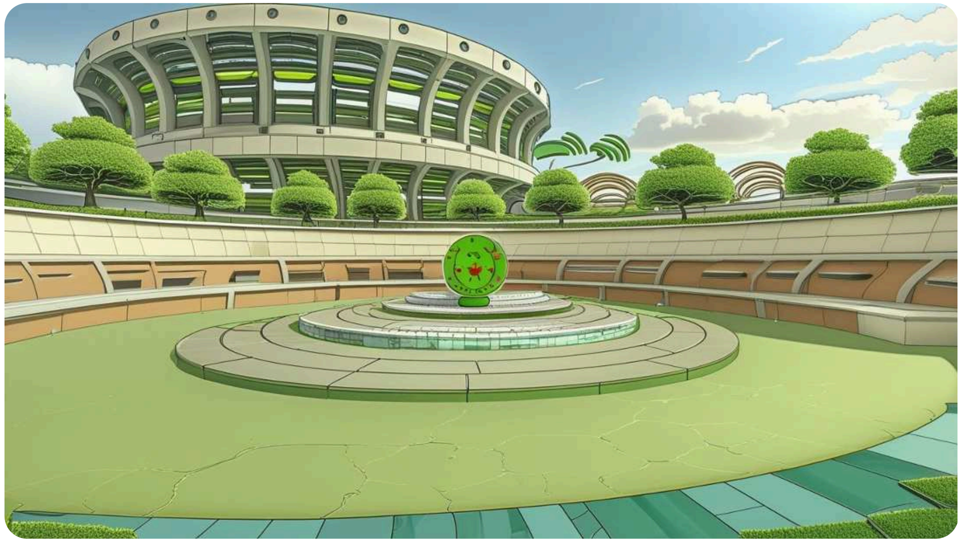
The ship in the desert

On 13 August 1868, a huge earthquake hit the city of Arica, which was part of Peru at that time. About an hour later, a tsunami about 16 metres high arrived. The waves picked up an American warship, the *USS Wateree*, and carried it about 430 metres onto the land! When the water went back, the ship was standing on the sand, far from the sea. The tsunami crossed the whole Pacific Ocean: it reached Hawaii and New Zealand about 15 hours later.

CHAPTER 12

W11 · Producto y audiencia

The Safety Fair



At the safety fair, you present your flyer to people who did not write it.

Our flyers are finished — now let's tell everybody!

Today is the safety fair. Parents, teachers and students from other classes walk around and look at the flyers. You stand next to yours, explain it and answer questions. The visitors didn't write your flyer, so they are the perfect test: can they understand it?

Now you try!

A **Your finished flyer.** ✋ Hand in · W10

Make the final copy of your flyer on A4 or A3 paper, with colours, three drawings and your corrections.

B **Two stars and a wish**

Write feedback for your partner: ⭐ ______ ⭐ ______ 🪄 ______

C **Fix the flyer.**

Find and correct five mistakes in Mateo's flyer (in the box above).

The tsunami of 1868 left a warship standing on the beach, far from the sea.

The earthquake is the alarm

When the earthquake is under the sea near Peru, the first wave can arrive in only 15 or 20 minutes. There is no time to wait for a siren. So if you are near the sea and the ground shakes so much that it is hard to stand up, **the earthquake is your alarm**. In 2001, a tsunami came to Camaná, in Arequipa, about 15 minutes after the earthquake. It was winter, so the beaches were almost empty, but the water went more than one kilometre inland.

The five steps that save you

1. 🌍 A strong earthquake near the sea is your alarm. Don't wait for a siren.
2. 🚶 Walk quickly away from the beach. Follow the green evacuation signs.
3. ⬆️ Go up: to a hill or a tall, strong building marked “Zona segura”.
4. ⏳ Stay there. More waves can come for many hours.
5. ✅ Go back only when the authorities say it is safe.

A Wave from Far Away

On 29 July 2025, a magnitude 8.8 earthquake happened in Kamchatka, in Russia, on the other side of the Pacific. Peru closed 65 ports and many beaches, just in case. The next day, small waves arrived in Callao. Nobody was hurt — because Peru was ready.

Photo: Gaboct · CC BY-SA 4.0

✅ Checklist

☐ My flyer has a BEFORE part.

☐ My flyer has a DURING part.

☐ My flyer has an AFTER part.

☐ I use must, must not or should.

☐ I say what the family must have ready.

☐ I say where to go.

☐ Every instruction starts with a verb.

☐ It has 90–110 words.

☐ Each part has a drawing.

Check the verbs

Instructions start with the verb: **Pack** your backpack. **Walk** to the meeting point. **Don't** run. Look at every sentence in your flyer. Does the verb come first? Is it clear?

Mateo's flyer — find the mistakes!

- BEFORE: You must not pack water.
- DURING: Run outside quickly!
- DURING: You should drop, cover and hold on.
- AFTER: Use the lift.
- AFTER: Go to the beach and look at the sea.

Read it out loud

At the safety fair, you will explain your flyer. Practise now: read it out loud so that someone at the back of the room can hear you. Speak slowly, stop at each full stop and look up at your listeners.

Making It Better

Peer check: a friend reads your flyer like the family in your street would.

I forgot the AFTER part! Where's my AFTER part?

Good writers always check their work. Swap your flyer with a partner and use the checklist. Then say **two stars and a wish**: two things you like, and one thing to make better.

Now you try!

A The five steps in order, in your notebook. ✋ Hand in · W4

Write the five steps in order. Use imperatives: *Go…, Walk…, Stay…, Don't…*

B Tsunami: true or false?

1. A tsunami is only one wave. **T / F**
2. After an earthquake near Peru, the first wave can come in 15–20 minutes. **T / F**
3. If the sea goes back, it is a good time to look for shells. **T / F**
4. The 1868 tsunami reached New Zealand. **T / F**
5. You should drive your car to the beach to see the waves. **T / F**

C Read the signs.

Look at the tsunami sign in the photo. Where does the arrow tell you to go? Why is it green?

D The ship in the desert — questions

1. When was the Arica tsunami? ______
2. How high was the wave? ______
3. How far did the Wateree travel onto the land? ______

E Role-play: on the beach

Student A is at the beach with a younger cousin. The ground shakes. Tell your cousin what to do in four instructions.

Now you try!

A Your first draft, with its three drawings. ✋ Hand in · W9

Write your first draft in the planner. Then draw one picture for each part. Count your words: ______

B Rule or advice?

Write **R** (rule: must) or **A** (advice: should).

1. ___ Drop, cover and hold on.
2. ___ Check your backpack every six months.
3. ___ Don't use the lift.
4. ___ Put a family photo in the backpack.

BEFORE	DURING	AFTER
Heading:	Heading:	Heading:
Instructions (must / must not):	Instructions (must / must not):	Instructions (must / must not):
Drawing:	Drawing:	Drawing:

Word count: ______ (90–110)

Word bank

Disasters earthquake · tsunami · landslide · flood · volcano · ash

Safety safe zone · drill · exit · meeting point · shelter · danger · Drop, cover, hold on

Backpack emergency kit · water · torch · whistle · blanket · first-aid kit · radio

Grammar must · must not · should · have to · before · during · after

Sentence starters

Before: You must pack… · Keep the backpack… · Find the safe zone… · You should check…
During: Drop, cover and hold on. · You must not run… · Stay away from…
After: Walk to the meeting point. · If you live near the sea, … · Call 119 to…

DID YOU KNOW?

Count your words: titles and headings count too. Aim for 90–110.

CHAPTER 5

Science link · Reading plan

Mountains of Fire

El Misti (5,822 m) stands above Arequipa, a city of more than a million people.

All of Peru's volcanoes are in the south. Look at the map!

A **volcano** is a mountain with an opening at the top. Deep under it, rock is so hot that it melts. This hot, liquid rock is called **magma**, and when it comes out it is called **lava**. When a volcano erupts, it can throw out lava, rocks, gas and clouds of grey **ash**.

Where the ocean plate sinks deep under the land, rock melts and rises as magma.

The Nazca Plate makes volcanoes too. When it goes very deep under Peru — about 100 kilometres down — it gets very hot and helps the rock above it to melt. The magma rises and makes volcanoes. Peru has 16 active volcanoes, and they are all in the south: in Arequipa, Moquegua, Tacna and Ayacucho. The IGP watches 13 of them every day.

The biggest eruption in South America

On 19 February 1600, the volcano **Huaynaputina**, in Moquegua, exploded. It was the biggest eruption in the history of South America. Ash fell like grey snow on the city of Arequipa, and about a dozen villages were buried. About 1,500 people died. The ash went so high into the sky that the next summer was very cold in Europe and Asia!

CHAPTER 10

W9 · Producto y audiencia

Writing My Flyer

Plan, write and draw: your flyer, your words.

My Flyer-Machine needs three parts, 90 to 110 words and a drawing in each part!

Now it's your turn. Your flyer has **three parts** — before, during and after an earthquake. Each part has a heading, at least one instruction and a drawing. Altogether it has **90 to 110 words**. Use **must**, **must not** and **should**.

D Good flyer or bad flyer?

Read this flyer text: *"Earthquakes are a natural phenomenon that has happened many times in the history of our country and it is very important that all the families think about them."* Rewrite it as two instructions for a real family.

An eruption can throw ash high into the sky and cover fields far away.

The White City

Arequipa is called the "White City" because many of its old churches and houses are built with *sillar*, a white stone made from volcanic ash. The volcano El Misti, 5,822 metres high, stands above the city.

Today, two volcanoes are very active. **Ubinas** is the most active volcano in Peru. **Sabancaya** started erupting in November 2016, and it is still erupting — it sends small clouds of ash into the sky almost every day.

DID YOU KNOW?

Sabancaya (5,960 m) is higher than El Misti (5,822 m) and Ubinas (5,672 m).

Now you try!

A

The three headings of your flyer with one instruction under each.

✋ Hand in · W8

Write your three headings and one instruction under each. Example: **BEFORE: Get ready!** — Pack your backpack.

B **Label the model flyer.**

Write the name of each part next to the model flyer: **title** · **heading** · **instruction** · **picture** · **caption**. How many instructions are there in total?

C **Find the hidden instruction.**

1. A whistle is loud. → ____________
2. Windows can break. → ____________
3. The meeting point is in the park. → ____________

Is your family ready for an earthquake?

BEFORE	DURING	AFTER
Get ready!	**Drop, cover, hold on!**	**Go to the meeting point**
🎒	🛡️	🚶
Pack an emergency backpack with water, a torch, a whistle and a first-aid kit. Keep it by the door. Find the safe zones in your house.	Don't run outside. Drop down, cover your head and hold on. Stay away from windows.	Walk out calmly and use the stairs, not the lift. If you live near the sea, go up at once. Call 119 to leave a message.
You must check the backpack every six months.	*The shaking can last for two minutes or more.*	*Our meeting point: the park on the corner.*

Emergency: 116 · 105 · 106 · Voice messages: 119

Model flyer — 114 words.

The parts of a flyer

🔠 **Title**	The big words at the top. It says what the flyer is about.
🏷️ **Heading**	The name of each part: BEFORE, DURING, AFTER.
👉 **Instruction**	It tells you what to do: Pack…, Drop…, Walk…
🖼️ **Picture**	It shows the action, so you understand without reading.
💬 **Caption**	A short sentence under a picture.

The hidden instruction

Sometimes a sentence doesn't look like an instruction, but it is one. A *whistle is loud and rescuers can hear it*. What does it really mean? **Pack a whistle!** On your flyer, it's better to write the instruction clearly: verb first.

Now you try!

A Label the volcano.

Draw a volcano and label: **crater · lava · magma · ash cloud**

B Compare the volcanoes.

Use **higher than** and **the highest**.

1. Sabancaya is ________ El Misti.
2. El Misti is ________ Ubinas.
3. ________ volcano is Sabancaya.

C True or false?

1. All of Peru's active volcanoes are in the south. **T / F**
2. Huaynaputina erupted in 1970. **T / F**
3. Sabancaya is still erupting. **T / F**
4. Arequipa is called the White City because of snow. **T / F**

D Experiment: make a volcano

With your teacher, make a volcano with baking soda, vinegar and red food colouring. Then write two safety rules with **must** and **must not**: *We must wear… We must not…*

CHAPTER 6

W5 · Final y entrega

When People Get Hurt

A street in Pisco after the earthquake of 15 August 2007. Rescue trucks bring help.

Photo: Senior Airman Shaun Emery, U.S. Air Force · Public domain (U.S. military)

Where does it hurt? Tell a teacher!

On 15 August 2007, at 6:40 in the evening, a strong earthquake shook the south coast of Peru. It was magnitude 8.0, and the ground shook for about two minutes or more. In Pisco, Ica and Chincha, thousands of old adobe houses fell down. The San Clemente church in Pisco fell during a Mass. 596 people died and more than 1,200 people were hurt.

CHAPTER 9

W8 · Producto y audiencia

How a Safety Flyer Works

A good sign or flyer works in seconds — like this tsunami evacuation sign in Barranco, Lima.

Photo: Gaboct · CC BY-SA 4.0

A family reads a flyer in ten seconds. So every word must work!

A **safety flyer** is a short page that tells people what to do. The family in your street won't read a long report. They will read your flyer quickly — maybe while the ground is moving! So a good flyer has a clear title, three parts with **headings**, short **instructions** and **pictures** with **captions**.

D The drill chant

When the ground begins to shake, / drop, cover, hold on, for goodness' sake! / No running, no shouting, no pushing, no fear, / walk to the meeting point — the teacher's here! Say it with actions.

E Before, during or after?

Sort these actions into three columns: *pack the backpack · drop, cover, hold on · go to the meeting point · learn the 119 number · stay away from windows · check if anyone is hurt · agree a family meeting point · stay away from the beach*

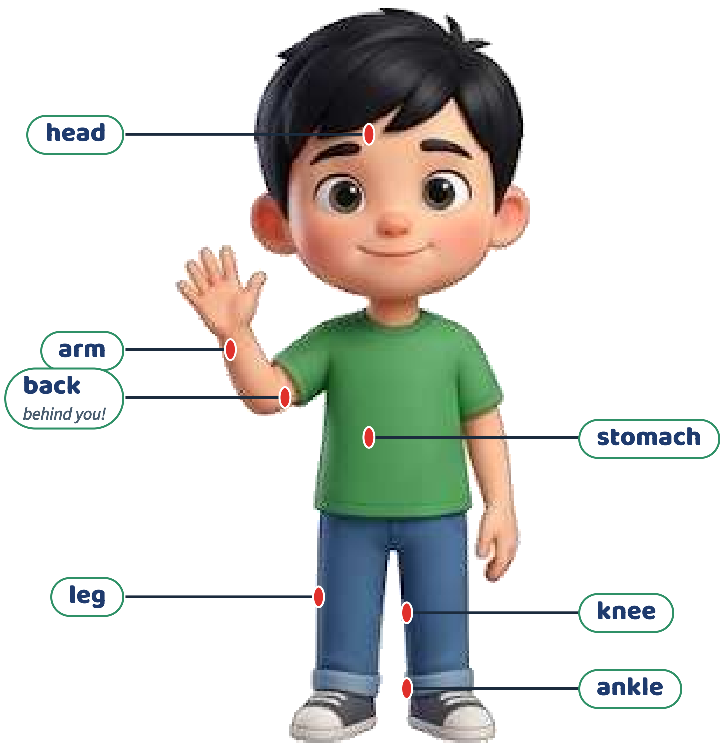
BEFORE	DURING	AFTER

Firefighters, police officers, doctors and ordinary people worked day and night to help. Planes brought food, water, tents and medicine to Lima, and trucks took them to Pisco. When people are hurt, everybody can help — even children. You can stay calm, help the person to stay calm, and tell an adult.

Relief supplies arrive at Lima airport for Pisco, August 2007.

Photo: Tech. Sgt. Jeremy Lock, U.S. Air Force · Public domain (U.S. military)

My body



Parts of the body. Your back is behind you — point to it!

When somebody is hurt, ask: **Where does it hurt?**
My **knee** hurts. · My **back** hurts. · I've got a **headache**. · I've got a **stomachache**. · I've got a **cut** on my arm. · I've got a **bruise** on my leg.

Who do you tell? What do you say?

Tell an adult at once: a teacher, a school brigade member, your mum or your dad. Say who is hurt, where they are and what happened: "Please come! Lucía is hurt. She's in the playground.

A

The map of your classroom with the safe zone and the way out.

✋ Hand in · W7

Draw a map of your classroom from above. Mark: the door, the windows, the safe zone (green) and an arrow for the way out.

B

Put the drill in order.

☐ The teacher counts everybody. ☐ The alarm sounds. ☐ We walk out in a line. ☐ We drop, cover and hold on. ☐ The shaking stops. ☐ We wait for instructions.

C

Game: Simon says

Play in groups. The leader gives drill instructions (*Drop! Cover your head! Walk to the door!*). Only follow them after "Simon says". The last player out becomes the leader.

practises for its own dangers: earthquakes, tsunamis, floods or volcanoes.

No corro, no grito, no empujo

Children in Peru learn three rules for evacuating: *No corro, no grito, no empujo* — I don't run, I don't shout, I don't push. Running and pushing make people fall; shouting makes people scared and stops them hearing the teacher.

She fell and her ankle hurts." If it is serious, the adult calls for help.

- **116** Firefighters *Bomberos*
- **105** Police *Policía*
- **106** Ambulance *SAMU*
- **115** Civil defence *Defensa Civil*
- **119** Leave a voice message *Mensajes de voz*

Line 119: Leave a Message

After a big earthquake, millions of people try to call their families, and the lines get busy. In Peru you can call **119** and record a short voice message. Your family calls 119 too and listens to it. Every family should know this number!

Sofía: Are you OK? Where does it hurt?

Mateo: My knee hurts. I fell on the stairs.

Sofía: Don't move. I'll get Miss Rosa.

Mateo: Thank you. Please be quick!

Now you try!

A A four-line dialogue: somebody is hurt and you help.

✋ Hand in · W5

Write your own four-line dialogue. Line 1: ask a question. Line 2: say where it hurts. Line 3: help. Line 4: finish.

B Label the body.

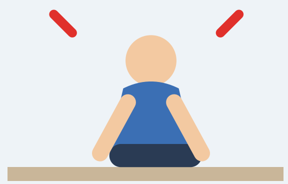
Write: **head** · **arm** · **leg** · **back** · **knee** · **ankle** on the picture.

C What's the problem?

Match the problem and the help.

1. I've got a cut.	a. Sit down and put something cold on it.
2. I've got a bruise.	b. Wash it and put a plaster on it.
3. My ankle hurts a lot.	c. Don't walk. Tell an adult.
4. I've got a headache.	d. Sit in the shade and drink water.

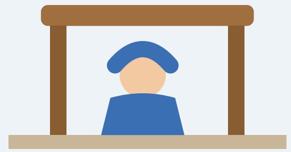
Drop, cover, hold on



DROP

Agáchate

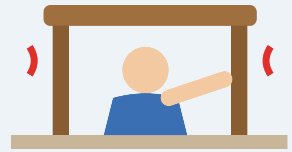
Drop onto your hands and knees.



COVER

Cúbrete

Cover your head and neck, or go under a strong table.



HOLD ON

Sujétate

Hold on until the shaking stops.

When the shaking starts, don't run outside. Stairs and doors are dangerous during the shaking. **Drop** down onto your hands and knees. **Cover** your head and neck with your arms, or go under a strong table. **Hold on** until the shaking stops. In Spanish, we say: *agáchate*, *cúbrete y sujétate*.

The three moments of a Peruvian drill

The three moments

1 🔔 1. The alarm sounds: drop, cover and hold on in the safe zone.

2 🚶 2. When the shaking stops, walk out calmly by the evacuation route: no corro, no grito, no empujo.

3 🙋 3. Go to the meeting point. The teacher counts everybody.

Several times a year, the whole country practises together in the **Simulacro Nacional Multipeligro** (national drill). In May 2026, more than nine million schoolchildren took part! Every region

The green "S" sign marks a safe zone inside a building.

Photo: AgainErick · CC BY-SA 3.0 (also GFDL)

D Emergency numbers quiz

Who do you call?

1. A fire: ______
2. An ambulance: ______
3. The police: ______
4. To leave a message for your family: ______

E Who, where, what?

Your friend Diana fell in the library and her arm hurts. Write what you say to the teacher (who, where, what).

CHAPTER 7

W6 · Producto y audiencia

The Backpack

🎒 The INDECI list (for two people)

- water (4 × ½ litre) — *agua embotellada*
- torch and batteries — *linterna y pilas*
- whistle — *silbato*
- tinned food — *comida enlatada*
- masks — *mascarillas*
- toilet paper — *papel higiénico*
- work gloves — *guantes*
- plastic bags — *bolsas*
- first-aid kit — *botiquín*
- small radio — *radio portátil*
- blanket — *manta polar*
- biscuits and chocolate — *galletas y chocolate*
- hand gel — *gel antibacterial*
- emergency phone numbers — *agenda con teléfonos*
- rope (7 m) — *cuerda*
- pencils and a thick pen — *útiles para escribir*

After a big earthquake, the lights may go out. There may be no water in the taps, and the shops may be closed. You might have to leave your house quickly. That's why every family in Peru should have an **emergency backpack** (*mochila para emergencias*) ready near the door.

CHAPTER 8

W7 · Producto y audiencia

The Safe Zone and the Drill

Students in Lima practise at the national drill (simulacro).

In every school in Peru there are green signs on the walls. They say **ZONA SEGURA EN CASO DE SISMOS** — safe zone in case of earthquakes. They show the safest places inside a building: next to strong columns, under strong beams and away from windows, shelves and things that can fall.

D Mateo's backpack

Mateo packed: a football, three comics, a torch with no batteries, a big bottle of cola and his tablet. Write four sentences telling him what he must and must not do.

E Take it home

Show the INDECI list to your family. Tick what you already have and circle what is missing. You will use this in Chapter 12.

The real INDECI rules

24 h
It is for the first 24 hours.

8 kg
It must not be heavier than 8 kilos.

1 : 2
One backpack for every two adults.

1 L
About one litre of water for each person.

The INDECI list (for two people)

- water (4 × ½ litre) — *agua embotellada*
- torch and batteries — *linterna y pilas*
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- pencils and a thick pen — *útiles para escribir*

must = it is a rule, it's very important. **must not** = don't do it!
You **must** pack water because the taps may not work.
You **must** keep the backpack near the door.
You **must not** make it too heavy.
You **should** check the food and water every few months. (**should** = it's a good idea)

Why a Whistle?

If you are trapped or lost, shouting makes you tired very quickly. A whistle is loud, light and easy to use, and rescuers can hear it from far away.

The Reserve Box

INDECI also recommends a **caja de reserva** (reserve box). It stays at home, in a safe, dry place, and has food and water for the family for days 2, 3 and 4.

Now you try!

A Your list of eight things, each with a reason. ✋ Hand in · W6

Write eight things and a reason: *We must pack a torch because the lights may go out.*

B must or must not?

1. You ______ pack water.
2. You ______ put a heavy computer in it.
3. You ______ keep it near the door.
4. You ______ forget the torch.

C Maths: how heavy?

Water 2 kg · food 1.5 kg · blanket 1 kg · torch and radio 1 kg · first-aid kit 0.5 kg. Total: ____ kg. Is it under 8 kg? ____